

Modeling Nicotinic Receptor Dynamics in Alzheimer's Disease

Valentin Jacquin under the supervision of Dr. Boris Gutkin
Group for Neural Theory
ENS Paris, France

Abstract

Alzheimer's disease (AD) is a prevalent neurodegenerative disorder characterized by progressive cognitive decline. During its early stage, the $A\beta$ molecule accumulates and provokes various disruptions in the neural system, notably leading to changes in prefrontal cortex (PFC) activity. Previous studies have explored the link between its disruption of nicotinic acetylcholine receptors (nAChRs) and alterations in PFC dynamics observed in early AD. Building upon their computational model of the PFC, we aim to investigate how nAChR dysfunction affects PFC activity during working memory tasks, which are known to be impaired in AD. Incorporating nAChR modulation, the model will allow us to explore the impact of receptor dysfunction on network bistability and transition dynamics. By implementing a ring attractor architecture, we will simulate working memory processes and assess how impairments in specific nAChR subtypes influence the network's ability to maintain stable activity states. We will analyze firing rates, transition frequencies, and working memory performance metrics to elucidate the relationship between nAChR dysfunction and cognitive deficits in AD.

Keywords: Alzheimer's disease, nicotinic acetylcholine receptors, prefrontal cortex, computational modeling, working memory.

Scientific Background and Context

Alzheimer's disease (AD) is a widespread neurodegenerative disorder with a high prevalence in elderly population, currently affecting over 50 millions individuals worldwide [Brejijeh and Karaman, 2020]. However, the origins of the cognitive impairments associated with AD remain poorly understood. While Palop et al. [2007] have reported increases in cortical activity in mice with AD, human imaging studies have predominantly observed hypofrontality in AD patients [Choi et al., 2023]. This apparent discrepancy highlights how the underlying mechanisms remain debated. Experimental evidence indicates that amyloid-beta ($A\beta$) peptides modulate neuron's signals by either activating or inhibiting nicotinic acetylcholine receptor (nAChR). More specifically, $\alpha 7$ and $\beta 2^*$ nAChR's subunits seems to be particularly affected by $A\beta$ interactions [Lombardo and Maskos, 2015, Parri et al., 2011].

In this work, we hypothesize that the progressive accumulation of $A\beta$ disrupts nAChR mechanisms, leading to a dysregulation in prefrontal cortex (PFC) network dynamics. Thus, contributing to the cognitive deficits observed in AD. As nAChRs are only expressed in interneurons in layer II/III [Poorthuis et al., 2013], nicotinic signaling is expected to impact pyramidal cells through interneurons modulation. In this regard,

Koukouli et al. [2017] demonstrated the link between nAChRs disruption and pyramidal neurons activity, underlying the interneurons implication in PFC network dynamics. The specific contribution of different receptor subunits to this modulation has also been highlighted [Maskos and Lombardo, 2014]. Still, the precise interactions between nAChR dysfunction and the resulting alterations in PFC network dynamics in the context of AD remain to be explained.

To better capture these mechanisms, previous work developed a reduced population-level model describing interactions between pyramidal neurons and major interneuron subtypes within the PFC [Koukouli et al., 2025]. This model has been fitted to their own produced data. It reproduces the hyperactivity observed in their genetically modified mice and identifies $\alpha 7$ nAChRs as critical modulators of this phenomenon. Moreover, it suggests that $\beta 2^*$ and $\alpha 5$ nAChRs could represent promising therapeutic targets for restoring physiological network balance.

Research Question and Working Hypotheses

While previous explorations have shown promising results in linking nAChR dysfunction to altered PFC activity patterns [Rooy et al., 2021, Koukouli et al., 2025] they primarily focused on resting-state dynamics. However, to link these findings to the observed cognitive deficits in AD, it is essential to investigate how these network alterations manifest during active cognitive tasks. The PFC is known to be a key region in working memory, which deteriorates in early AD [Huntley and Howard, 2010]. Furthermore, electrophysiological recordings during working memory tasks have been shown to reliably distinguish AD patients from healthy controls [Perpetuini et al., 2020], underscoring the connection between altered neural dynamics and cognitive impairment.

Therefore, working memory task provides a good candidate to explore how nAChR dysfunction affects PFC network dynamics in an active context. We will investigate how disruptions in the different nAChR subtypes influence the network's ability to switch between two self-sustained states, which are thought to underlie working memory processes [Compte et al., 2003]. We hypothesize that impairments in specific nAChR subtypes will lead to deficits in the stability and transition dynamics between these states, thereby contributing to the working memory impairments observed in AD.

To another extent, we will also explore how varying levels of external input or sensitivity on specific nAChR subtypes affect these dynamics, building on previous findings that identified $\beta 2^*$ and $\alpha 5$ nAChRs as key modulators of PFC activity [Koukouli et al., 2025].

Planned Methodology

To tackle this question, we will build upon the previously developed nAChR-modulated PFC network model described in [Koukoulis et al. \[2025\]](#).

Modulation of parameters affecting bistability

Previous results have shown an oscillation between high and low activity states in the PFC [\[Rooy et al., 2021\]](#), but at another order of magnitude than the one observed experimentally during working memory tasks [\[Compte et al., 2003\]](#). Thus, it is needed to bridge the gap closer to experimental observations, as we aim to maintain a biologically relevant model. To do so, we will modulate current inputs as we expect them to highly affect the characteristics of both states.

Working memory architecture

Once we achieve a network behavior consistent with experimental findings, we will build a ring attractor network. In this architecture, units are arranged along a circular topology where local excitation and global inhibition produce a localized bump of activity that can be sustained after stimulus offset, encoding a continuous variable such as spatial location. This architecture has been widely used to model working memory processes, notably in [\[Wang et al., 2004\]](#) where the authors also model the role of interneurons. However, they first do not consider nAChR modulation, and second use an arbitrary connectivity framework between nodes.

Here, we will implement a similar architecture but defining each node as an identical local PFC circuit described by our nAChR-modulated model. We will consider PYR to PYR connections as the driver of the excitatory connectivity, while PV interneurons will mediate the inhibitory connections between nodes, as they are both known to be able to project horizontally [\[Levitt et al., 1993, Druga et al., 2023\]](#). SOM and VIP interneurons will be considered as local modulators of the network dynamics as they mainly project locally [\[Karnani et al., 2016, Riedemann, 2019\]](#).

Expected Observables and Data

We will primarily focus on analyzing the firing rates of the different neuron populations (PYR, PV, SOM, VIP) within the PFC network model. The following observables will be used to assess the impact of nAChR dysfunction and input modulation on network dynamics.

Characteristics of activity states

As previously mentioned, we first aim to get closer to experimental observations during working memory tasks [\[Compte et al., 2003\]](#). We will quantify the mean firing rates and variance of both high and low-activity states along to varying input current levels. We expect that increasing input current will lead to higher firing rates in both states.

Transition dynamics

We will quantify the frequency of transitions between high and low-activity states. This should be highly modulated by the input current, and we expect that stronger inputs may lead to prolonged high-activity states, potentially resulting in a one-stability state at a certain threshold. Whereas, lower inputs may favor a more stable low-activity state.

Working memory performance metrics

In the ring attractor model, we will first assess the network's ability to form and maintain a bump of activity in response to a transient stimulus. Then, we will measure the population vectors to quantify the accuracy of working memory representations. We expect the network to be able to maintain stable activity bumps, successfully encoding the stimulus information. In the context of nAChR dysfunction, we will evaluate how impairments in specific receptor subtypes affect the stability and fidelity of these representations.

References

- Zeinab Breijyeh and Rafik Karaman. Comprehensive review on alzheimer's disease: Causes and treatment. *Molecules*, 2020. doi: 10.3390/molecules25245789.
- Jungmi Choi, Boncho Ku, Dieu Ni Thi Doan, Junwoo Park, Wonseok Cha, Jaeuk U. Kim, and Kun Ho Lee. Prefrontal eeg slowing, synchronization, and erp peak latency in association with predementia stages of alzheimer's disease. *Frontiers in Aging Neuroscience*, 2023. doi: 10.3389/fnagi.2023.1131857.
- Albert Compte, Christos Constantinidis, Jesper Tegnér, Sridhar Raghavachari, Matthew V. Chafee, Patricia S. Goldman-Rakic, and Xiao-Jing Wang. Temporally irregular mnemonic persistent activity in prefrontal neurons of monkeys during a delayed response task. *Journal of Neurophysiology*, 2003. doi: 10.1152/jn.00949.2002.
- Rastislav Druga, Martin Salaj, and Azzat Al-Redouan. Parvalbumin - positive neurons in the neocortex: A review. *Physiological Research*, 2023. doi: 10.33549/physiolres.935005.
- J.D. Huntley and R.J. Howard. Working memory in early alzheimer's disease: a neuropsychological review. *International Journal of Geriatric Psychiatry*, 2010.
- Mahesh M. Karnani, Jesse Jackson, Inbal Ayzenshtat, et al. Opening holes in the blanket of inhibition: Localized lateral disinhibition by vip interneurons. *Journal of Neuroscience*, 2016. doi: 10.1523/JNEUROSCI.3646-15.2016.
- F. Koukouli et al. Nicotine reverses hypofrontality in animal models of addiction and schizophrenia. *Nature Medicine*, 2017.
- F. Koukouli et al. The alpha7 nicotinic acetylcholine receptor mediates network dysfunction in a mouse model of local amyloid pathology. *Molecular Psychiatry*, 2025.
- Jonathan B. Levitt, David A. Lewis, Takashi Yoshioka, and Jennifer S. Lund. Topography of pyramidal neuron intrinsic connections in macaque monkey prefrontal cortex (areas 9 and 46). *Journal of Comparative Neurology*, 1993. doi: 10.1002/cne.903380304.
- S. Lombardo and U. Maskos. Role of the nicotinic acetylcholine receptor in alzheimer's disease pathology and treatment. *Neuropharmacology*, 2015.
- U. Maskos and S. Lombardo. Nicotinic receptors and alzheimer's disease. *Progress in Neurobiology*, 2014.
- JJ Palop et al. Aberrant excitatory neuronal activity and compensatory remodeling of inhibitory hippocampal circuits in mouse models of alzheimer's disease. *Neuron*, 2007.
- H.R. Parri, C.M. Hernandez, and K.T. Dineley. Research update: Alpha7 nicotinic acetylcholine receptor mechanisms in alzheimer's disease. *Biochemical Pharmacology*, 2011.
- D. Perpetuini et al. Working memory decline in alzheimer's disease is detected by complexity analysis of multimodal eeg-fnirs. *Entropy*, 2020.
- Rogier B. Poorthuis, Bernard Bloem, Benita Schak, et al. Layer-specific modulation of the prefrontal cortex by nicotinic acetylcholine receptors. *Cerebral Cortex*, 2013. doi: 10.1093/cercor/bhr390.
- Therese Riedemann. Diversity and function of somatostatin-expressing interneurons in the cerebral cortex. *International Journal of Molecular Sciences*, 2019. doi: 10.3390/ijms20122952.
- Marie Rooy, Ivan Lazarevich, Fani Koukouli, Uwe Maskos, and Boris Gutkin. Cholinergic modulation of hierarchical inhibitory control over cortical resting state dynamics: Local circuit modeling of schizophrenia-related hypofrontality. *Current Research in Neurobiology*, 2021. doi: <https://doi.org/10.1016/j.crneur.2021.100018>.
- X-J Wang, J Tegnér, C Constantinidis, and P S Goldman-Rakic. Division of labor among distinct subtypes of inhibitory neurons in a cortical microcircuit of working memory. *Proceedings of the National Academy of Sciences*, 2004. doi: 10.1073/pnas.0305337101.